

8-amaliy mashg'ulot. Diffuziya modellari

2026–2027 o'quv yili · 30 ta topshiriq

Mavzu. DDPM oldinga jarayoni va $q(x_t | x_0)$ yopiq formulasi, noise schedule (chiziqli, cosine) va SNR, posterior $q(x_{t-1} | x_t, x_0)$, ε -, x_0 - va v -prediction, ELBO va $\mathcal{L}_{\text{simple}}$, score function va Tweedie formulasi, DDPM va DDIM sampling, DDIM inversion, classifier-free guidance, vaqt kodlash (time embedding), U-Net va latent diffusion, SDEdit, baholash (FID, MMD).

Maqsadlar. Talaba: (1) $\bar{\alpha}_t$, SNR, posterior va sampler qadamlarini qo'lda hisoblaydi va formulalarni keltirib chiqaradi; (2) analitik “ideal tarmoq” ε^* bilan samplerlarni tekshiradi; (3) 2D ma'lumotda (**make_moons**) kichik DDPM ni o'qitadi, DDIM, classifier-free guidance, SDEdit va inversiyani qo'llaydi va natijani son bilan baholaydi.

Baholash (100 ball). A bo'lim (1–12): $12 \times 2 = 24$; B bo'lim (13–16): $4 \times 1,5 = 6$; C bo'lim (17–30): $14 \times 5 = 70$. Kodli topshiriqda: to'g'ri va takrorlanuvchi natija — 3 ball, kod sifati (imzo, urug', indekslash $t \leftrightarrow t-1$, `torch.no_grad()`) — 1 ball, izoh — 1 ball. Topshirish: **solutions.py** (har kodli topshiriq — alohida funksiya va **main**) va hisobot (PDF).

Muhitni sozlash (Python 3.11 yoki 3.12, faqat CPU)

```
python -m venv .venv; source .venv/bin/activate (Windows: .venv\Scripts\activate); pip
install -r requirements.txt. Versiyalar qat'iy belgilangan: numpy==2.4.4, scipy==1.17.1,
pandas==3.0.2, scikit-learn==1.8.0, scikit-image==0.26.0, matplotlib==3.10.9, pillow==12.2.0,
opencv-python-headless==4.13.0.92, torch==2.14.0, torchvision==0.29.0. Ishga tushirish: python -W
error::DeprecationWarning solutions.py — ogohlantirishsiz. Ma'lumotlar faqat sintetik (1D Gauss
aralashmasi, sklearn.datasets.make_moons) — internet kerak emas. Urug' 42, torch.set_num_threads(1);
umumiy runtime 3 daqiqadan kam.
```

Belgilashlar. $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{s \leq t} \alpha_s$, $\text{SNR}(t) = \bar{\alpha}_t / (1 - \bar{\alpha}_t)$; chiziqli jadval: β_t 10^{-4} dan 0,02 gacha teng qadamli, $T = 1000$. Kodda indeks 0 dan: `ab[t-1] = \bar{\alpha}_t`.

A. Qo'lda hisoblash va keltirib chiqarish (1–12)

- $\bar{\alpha}_t$ va SNR.** $T = 4$, $\beta = (0,05; 0,1; 0,15; 0,2)$. $\bar{\alpha}_1, \dots, \bar{\alpha}_4$ ni, $q(x_4 | x_0)$ ning o'rtacha koeffitsiyenti va dispersiyasini, $t = 2$ va $t = 4$ da SNR ni (dB) toping.
- Yopiq formula.** $q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$ ni induksiya bilan isbotlang. $\bar{\alpha}_t = 0,36$, $x_0 = -0,5$, $\varepsilon = 0,5$ uchun x_t ni toping.
- Chiziqli jadval.** $\beta_t = \beta_1 + (t-1)\Delta$, $\Delta = (\beta_T - \beta_1)/(T-1)$. β_{501} ni toping. $\ln \bar{\alpha}_t = \sum \ln(1 - \beta_s) \approx -\sum \beta_s$ yaqinlashishida $\sum_{s \leq t} \beta_s = t\beta_1 + \Delta t(t-1)/2$ ni isbotlang va $t = 251$ da $\bar{\alpha}_t$ ni baholang (aniq qiymat: 0,521423). SNR necha dB?
- Cosine jadval.** $\bar{\alpha}_t = g(t)/g(0)$, $g(t) = \cos^2(\frac{t/T+s}{1+s} \frac{\pi}{2})$, $s = 0,008$, $T = 1000$: $\bar{\alpha}_{250}, \bar{\alpha}_{500}, \bar{\alpha}_{750}$ ni hisoblang. Nega $\beta_t = \min(1 - \bar{\alpha}_t/\bar{\alpha}_{t-1}, 0,999)$ qirqish kerak ($t = T$ da nima bo'ladi)?
- Nega $\bar{\alpha}_T \approx 0$.** O'zgarmas $\beta = 0,02$ da $\bar{\alpha}_T \leq 10^{-3}$ uchun eng kichik T ? Chiziqli jadval $T = 200$ da ($\beta: 10^{-4} \rightarrow 0,02$) $\bar{\alpha}_T = 0,1322$: x_T da x_0 ning koeffitsiyenti qancha va bu sampling uchun nega xavfli?
- Posterior.** $T = 2$, $\beta_1 = \beta_2 = 0,2$. $\beta_2, \tilde{\mu}_2$ ning x_0 va x_2 koeffitsiyentlarini toping; $x_0 = 1$, $x_2 = 0,5$ da $\tilde{\mu}_2$ ni hisoblang.
- ε -parametrlash.** $\tilde{\mu}_t = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\varepsilon)$ ni keltirib chiqaring. 6-topshiriq ma'lumotlarida ε ni toping va shu formula bilan $\tilde{\mu}_2$ ni tekshiring.
- v -prediction.** $v = \sqrt{\bar{\alpha}_t}\varepsilon - \sqrt{1-\bar{\alpha}_t}x_0$ uchun $x_0 = \sqrt{\bar{\alpha}_t}x_t - \sqrt{1-\bar{\alpha}_t}v$ va $\varepsilon = \sqrt{1-\bar{\alpha}_t}x_t + \sqrt{\bar{\alpha}_t}v$ ni isbotlang. $\bar{\alpha}_t = 0,64$, $x_0 = 0,5$, $\varepsilon = 1$: x_t, v va ikkala tiklashni tekshiring. $\|x_0 - \hat{x}_0\|^2$ ni $\|\varepsilon - \hat{\varepsilon}\|^2$ orqali ifodalang.
- Tweedie (Gauss ma'lumot).** $x_0 \sim \mathcal{N}(m, s^2)$ (1D). $\mathbb{E}[x_0 | x_t]$, optimal $\varepsilon^*(x_t)$, score va minimal yo'qotish $\mathbb{E}(\varepsilon - \varepsilon^*)^2 = \frac{\bar{\alpha}_t s^2}{\bar{\alpha}_t s^2 + 1 - \bar{\alpha}_t}$ ni keltirib chiqaring. $m = 1$, $s = 0,5$, $\bar{\alpha}_t = 0,36$, $x_t = 0,2$ da hisoblang.
- DDPM qadami.** $\beta_t = 0,02$, $\bar{\alpha}_t = 0,3$, $x_t = 0,4$, $\hat{\varepsilon} = 0,5$, $\sigma_t^2 = \beta_t$, $z = -1$: μ_θ va x_{t-1} ni toping.
- DDIM qadami.** $\bar{\alpha}_{\tau_i} = 0,3$, $\bar{\alpha}_{\tau_{i+1}} = 0,8$, $x = 0,4$, $\hat{\varepsilon} = 0,5$. $\eta = 0$ da $\hat{x}_0$ va yangi x ni toping; natija $q(x_{\tau_{i+1}} | \hat{x}_0)$ dan o'sha $\hat{\varepsilon}$ bilan olingan nuqta ekanini ko'rsating. $\eta = 1$ da σ_i va $\hat{\varepsilon}$ oldidagi koeffitsiyent qancha?
- CFG, vaqt kodlash, latent.** (a) $\varepsilon_\emptyset = (0,1; -0,2)$, $\varepsilon_c = (0,4; 0,1)$: $s = 0, 1, 3$ da $\tilde{\varepsilon}$. (b) $h = 2$, $\omega = (1; 0,01)$, $t = 100$: $\text{emb}(t) = (\sin \omega t, \cos \omega t)$. (c) $768 \times 768 \times 3$ rasm, $f = 8$, 4 kanalli latent: sonlar soni va siqish. (d) DDIM 25 qadam + CFG: bitta rasm uchun U-Net o'tishlari soni.

B. Konseptual savollar (13–16)

13. **Tipik xatolar.** Uch xatoning har biri nimaga olib keladi: (a) $\bar{\alpha}_T$ nolga yaqin emas; (b) rasm $[0, 1]$ da qoldirilgan; (c) tarmoqqa t berilmagan. Har biriga tuzatish yozing.
14. **DDIM.** Nega DDIM uchun qayta o'qitish kerak emas? Deterministik sampling qanday imkoniyat (inversiya) beradi va qadamlar kamaytirilganda nima yo'qotiladi?
15. **Guidance.** $s = 0$, $s = 1$, $s > 1$ nimani anglatadi? $p(x_t)p(c | x_t)^s$ talqini orqali nega s katta bo'lsa xilma-xillik kamayadi va juda katta s nega sifatni buzishi mumkin?
16. **Baholash.** 1D FID: $\mathcal{N}(1; 0.5^2)$ va $\mathcal{N}(0; 1)$. Nega piksel statistikasi (o'rtacha, std) yetarli emas? FID nega namunalar soniga bog'liq va o'quv to'plamini aynan ko'chirib olgan (memorization) modelni nega jazolamaydi — qanday qo'shimcha tekshiruv kerak?

C. Kodli topshiriqlar (17–30)

Umumiy kod (ko'chirish); har topshiriq — `task<N>_...()` funksiyasi, chiqish `[N]` bilan boshlanadi.

```
import math, random, numpy as np, torch, torch.nn as nn, torch.nn.functional as F
SEED = 42; torch.set_num_threads(1)
def set_seed(s=SEED): random.seed(s); np.random.seed(s); torch.manual_seed(s)
def linear_schedule(T=1000, b1=1e-4, bT=0.02): #  $\beta_t[t-1] = \beta_t$ 
    beta = np.linspace(b1, bT, T); return beta, 1 - beta, np.cumprod(1 - beta)
MU, S0 = np.array([-2.0, 2.0]), 0.3 # 1D:  $0.5 \mathcal{N}(-2, 0.3^2) + 0.5 \mathcal{N}(2, 0.3^2)$ 
def sample_mixture(rng, n): return rng.choice(MU, n) + S0 * rng.standard_normal(n)
def time_embedding(t, dim=32, max_period=10000.0): #  $[\sin(w_k t), \cos(w_k t)]$ 
    half = dim // 2; w = max_period * (-torch.arange(half, dtype=torch.float32) / half)
    arg = t.float()[ :, None] * w[None, :]; return torch.cat([torch.sin(arg), torch.cos(arg)], 1)
class EpsMLP(nn.Module): #  $\epsilon_{\theta}(x_t, t, c)$ 
    def __init__(self, n_classes=0, hidden=128, emb=32):
        super().__init__(); self.emb = emb
        self.cls = nn.Embedding(n_classes + 1, emb) if n_classes > 0 else None # oxirgisi = null
        self.net = nn.Sequential(nn.Linear(2 + emb, hidden), nn.SiLU(), nn.Linear(hidden, hidden),
                                nn.SiLU(), nn.Linear(hidden, hidden), nn.SiLU(), nn.Linear(hidden, 2))
    def forward(self, x, t, c=None):
        h = time_embedding(t, self.emb) + (self.cls(c) if self.cls is not None else 0)
        return self.net(torch.cat([x, h], 1))
```

2D DDPM protokoli (22–30): $T = 100$, $\beta = \text{torch.linspace}(1e-3, 0.2, 100)$ (float64 da `cumprod`, so'ng float32); ma'lumot `make_moons(n_samples=4000, noise=0.05, random_state=42)`, o'z o'rtacha/std si bilan standartlashtirilgan. O'qitish: `set_seed()`, `gen=torch.Generator().manual_seed(42)`, 3000 qadam, har qadamda tartib bilan: indekslar `randint(0,n,(256,))`, t `randint(1,101,(256,))`, ϵ `randn`, (shartli modelda) `rand(256)<0.1` bo'lsa `sinf ← null`; AdamW(`lr=2e-3`, `weight_decay=0`) + CosineAnnealingLR(`T_max=3000`); yo'qotish `F.mse_loss(eps_theta, eps)`. DDPM sampling: $\sigma_t^2 = \beta_t$; DDIM: $\tau = \text{np.linspace}(T, 1, S).round()$, $\eta = 0$, oxirgi qadamda $x = \hat{x}_0$.

17. **Jadvallar.** `task17_schedules()`: chiziqli va cosine ($s = 0.008$, qirg'ilgan β ning `cumprod` i) jadvallar uchun $t \in \{1, 126, 251, 501, 751, 1000\}$ da $\bar{\alpha}_t$, $\sqrt{\bar{\alpha}_t}$, SNR (dB). SNR birinchi marta < 0 dB bo'ladigan t ; β_{501} ; $\bar{\alpha}_T$ ($T = 1000$ va $T = 200$).
18. **Yopiq formula tekshiruvi.** `task18_closed_form()`: `rng=default_rng(42)`, 400 000 ta aralashma namunasi; 501 qadam ketma-ket shovqinlash va yopiq formula bilan x_{501} ; ikkala o'rtacha/std va nazariy std $\sqrt{\bar{\alpha}_t(4 + 0.3^2) + 1 - \bar{\alpha}_t}$.
19. **Posterior Monte Carlo.** `task19_posterior_mc()`: $t = 501$, $x_0 = 0.8$, $2 \cdot 10^6$ juftlik (x_{500}, x_{501}) oldinga jarayondan. $|x_{501} - v| < 0.01$ shartida ($v = -0.25702; 0.5$) x_{500} ning empirik o'rtacha, standart xato (SE) va dispersiyasini $\hat{\mu}_t$ (ikki shakl) va $\hat{\beta}_t$ bilan solishtiring.
20. **Ideal tarmoq bilan DDPM.** `task20_ddpm_1d()`: aralashma uchun $\mathbb{E}[x_0 | x_t]$ ni (komponent vaznlari $\times$ 9-topshiriqdagi Gauss posteriori) va $\epsilon^* = (x_t - \sqrt{\bar{\alpha}_t}\mathbb{E}[x_0 | x_t])/\sqrt{1 - \bar{\alpha}_t}$ ni yozing. 1000 qadamli DDPM, 20 000 namuna: $P(x > 0)$, $\mathbb{E}|x|$, $\text{std}|x|$ va haqiqiy aralashma bilan solishtirish.
21. **DDIM qadamlar soni.** `task21_ddim_1d()`: bitta `z=rng.standard_normal(20000)`, $S \in \{5, 10, 20, 50\}$, `ts=np.linspace(T-1, 0, S).round()` (0-indeks). O'sha statistikalar; bir xil z bilan ikki yurish farqi (determinizm).
22. **2D DDPM o'qitish.** `task22_train_ddpm_2d()`: yuqoridagi protokol. Parametrlar soni, $\bar{\alpha}_1$, $\bar{\alpha}_T$, har 500 qadamdagi o'rtacha loss. Nega loss nolga tushmaydi?
23. **Namuna sifati.** `task23_sample_quality()`: test — `make_moons(4000, noise=0.05, random_state=43)` ning birinchi 1000 tasi (o'quv statistikasi bilan standartlashtirilgan). 1000 ta namuna: DDPM-100, DDIM-20, DDIM-5 (bitta `gen`, ketma-ket), $\mathcal{N}(0, I)$ bazaviy. RBF-yadroli MMD² (kenglik 0.5, siljigan baho)

- va `scipy.spatial.cKDTree` bilan o'quv to'plamigacha eng yaqin qo'shni masofasining medianasi; tarmoq o'tishlari soni.
24. **Classifier-free guidance.** `task24_cfg(): make_moons(4000, noise=0.2, random_state=42)` (sinflar qisman ustma-ust), `EpsMLP(n_classes=2)`, $p_{\text{uncond}} = 0,1$. Har $s \in \{0; 1; 3; 7,5\}$ uchun `gen.manual_seed(42)` dan $500 + 500$ namuna ($c = 0$, $c = 1$). Hakam — `KNeighborsClassifier(25)` (haqiqiy ma'lumotda): aniqlik, $P(c | x)$ o'rtachasi, sinf ichidagi std, NN-masofa medianasi.
 25. **Vaqt kodlash.** `task25_time_embedding():` $\omega = (1; 0,01)$, $t = 10$ uchun emb; dim=32 da (50, 51), (50, 60), (50, 100), (1, 100) juftlari kosinus o'xshashligi; $\|\text{emb}(t)\|^2$.
 26. **Loss profili va ELBO vaznlari.** `task26_loss_profile(model):` o'qitilgan modelda $t \in \{1, 5, 10, 25, 50, 75, 100\}$ uchun 2000 o'quv nuqtasida MSE va $\lambda_t = \beta_t / (2\alpha_t(1 - \bar{\alpha}_t))$; $T = 1000$ chiziqli jadval uchun $\lambda_{1,11,101,501,1000}$.
 27. **Tweedie tekshiruv.** `task27_tweedie():` $t = 501$, $4 \cdot 10^6$ juftlik (x_0, x_t) ; $|x_t - v| < 0,01$ ($v = -1; 0; 0,5; 1$) da x_0 o'rtachasi (SE bilan), analitik $\mathbb{E}[x_0 | x_t]$ va score $-\varepsilon^* / \sqrt{1 - \bar{\alpha}_t}$.
 28. **Kamaymas yo'qotish.** `task28_irreducible_loss():` $t \in \{1, 51, 251, 501, 1000\}$, 400 000 namuna; $\mathbb{E}(\varepsilon - \varepsilon^*)^2$ ni aralashma uchun va 9-topshiriq formulasi bilan ($\text{Var } x_0 = 4,09$ li Gauss) solishtiring.
 29. **SDEdit.** `task29_sedit(model):` $[-2, 2]^2$ dagi 20×20 to'r nuqtalari; $t_0 \in \{10, 30, 60\}$ gacha shovqinlash va t_0 dan DDPM teskari jarayoni (har t_0 oldidan `gen.manual_seed(42)`). NN-masofa medianasi va o'rtacha siljish $\|x_{\text{out}} - x_{\text{in}}\|$. "Strength" t_0/T ning ma'nosini izohlang.
 30. **DDIM inversiyasi.** `task30_ddim_inversion(model):` 500 ta o'quv nuqtasini deterministik DDIM ODE bilan $x_0 \rightarrow x_T$ (`np.linspace(1,T,S)` bo'yicha) va qaytib $x_T \rightarrow x_0$; $S \in \{10, 50, 100\}$ uchun o'rtacha tiklash xatosi, std(z_T) va tarmoq o'tishlari soni.

Hujjatlar (documentation)

- <https://docs.pytorch.org/docs/stable/generated/torch.nn.Embedding.html>, <https://docs.pytorch.org/docs/stable/generated/torch.nn.SiLU.html>
- <https://docs.pytorch.org/docs/stable/generated/torch.optim.AdamW.html>, https://docs.pytorch.org/docs/stable/generated/torch.optim.lr_scheduler.CosineAnnealingLR.html
- <https://docs.pytorch.org/docs/stable/generated/torch.Generator.html>, <https://docs.pytorch.org/docs/stable/generated/torch.cumprod.html>
- https://scikit-learn.org/stable/modules/generated/sklearn.datasets.make_moons.html
- <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>
- <https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.cKDTree.html>
- <https://numpy.org/doc/stable/reference/random/generator.html>
- Ho, Jain, Abbeel (DDPM): <https://arxiv.org/abs/2006.11239>; Song, Meng, Ermon (DDIM): <https://arxiv.org/abs/2010.02502>; Ho, Salimans (CFG): <https://arxiv.org/abs/2207.12598>; Nichol, Dhariwal (cosine): <https://arxiv.org/abs/2102.09672>